# LECTURE 2: MEASUREMENT CONCEPTS

**Introduction**

Welcome to lecture 2. In this lecture we are going to explain the concept of measurement and see what it clearly means, we are also going to discuss the levels of measurement and their applicability and finally look at the relationships between the different scales of measurement. let us begin by looking at the learning outcomes for this session.

**Learning outcomes**

After this lecture, the learner should be able to:

a. Explain the concept of measurement.
b. List the attributes of scale of measurement.
c. Discuss the relationship between the measurement scales.

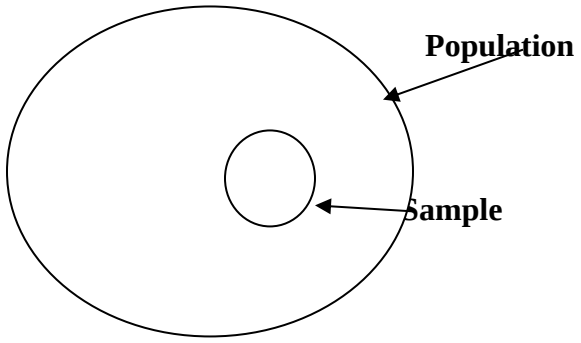
Differentiate between qualitative and quantitative data

In this course there are certain terms that are frequently used. These include:

## TEST AND TESTING

### *What is a test?*

A test is a device by which we can use **to sample** the candidate/students behavior. In this case behavior is performance.



- Testing is always a matter of sampling. We can never ask all the questions we could like to ask in test.
- The common kind of test that the teacher is used to is the paper and pencil (paper and pen) test, i.e. that in which the pupil is required to write or mark his / her answers on a paper. However, tests may take various other forms. In some cases, the pupil may indicate his answer orally (oral tests), in other he / she may be required to carry out certain activities during which he / she is observed and scored by an observer.
- A classroom test must be in harmony with ***instructional objectives and subject content.*** What are the instructional objectives? These are the learning objectives or what the course can do. For each topic, the teacher provides what the learners are expected / able to learn at the end of the topic presentation. These learning objectives are stated in **action-verbs (measurable and observable)** e.g. to identify, describe, discuss, explain, demonstrate, analyze, etc. (To be sure that these learning objectives are achieved, the preparation of a test should follow a systematic procedure. We will discuss this procedure later in this course.

# TYPE OF TESTS

Which is your favorite quotation in the bible? Bible is a document with so many tests. Our earthly tests include but not limited to:

## Achievement Tests (*Example: KCPE and KCSE)*

Achievement tests measure the correct status of learners with respect to proficiency in given area of knowledge or skill and hence can be used for selection. They are developed by experts. Achievement test can be:

- Standardized test (which are designed to cover content which are common to many classes of the same type of learners).
- Locally-developed (designed to measure particular set of learning outcomes, set by specific teacher. These are also called teacher made test-intended by teachers to test the areas they covered in class.

## Aptitude Tests

Aptitude tests are measures of ***potential.*** They are used to predict how well someone is likely to perform in future. Tests of general aptitude are variously referred to as scholastic aptitude tests, intelligence tests, and tests of general mental ability. Aptitude tests are also available to predict a person's likely level of performance after following some specific future instruction or training. Aptitude tests are standardized.

***Example:*** When Kenya Science Teachers College was established the only institution for training secondary school science teachers then it used aptitude test to selection the trainees.

## Personality Tests

Personality tests for diagnosis of behavior problems. They are designed to measure characteristics of individuals along a number of dimensions including:

- Attitudes towards self and others
- Team building

Most of these tests are self-report measures where an individual is asked to respond to a series of questions or statements and are available online.

## Standardized Tests (*Example: KCPE and KCSE, KNEC)*

A standardized test is one that is developed by experts or subject matter, specialists, administered interpreted using uniform procedures and performance standards (norm-referenced and criterion-referenced standards). That standardization here implies:

- Uniform setting standards
- Tests developed by subject matter specialists
- Uniform administration standards
- Uniform scoring procedure

Most standardized tests are objective, written tests requiring written responses.

## Locally-Developed Test (Teacher-Made Tests)

The opposite of standardized test is non- standardized test. Such tests are usually developed locally for a specific purpose. The tests are used by teachers in the classroom are examples of locally developed tests. Such test may:

- Be as good as a standardized test, but not often,
- Be more practical and more appropriate
- More likely to reflect what was taught in class to a greater degree than standardized tests

https://www.youtube.com/watch?v=em5KHBQlIAQ

## Assessment

***Assessment implies / involves getting information***. It is getting information on a learner's ability to demonstrate acquisition and application of outcomes of a course/program of learning, leading to the award of a qualification or transition to the next grade. E.G. reading ability, reading ability, and numeracy.

Assessors are expected to answer questions such as "***Is John making progress? "Is Jane a good learner?"*** In school settings or classroom situation, we get information about the learners through the following modes / techniques/methods:

Formal vs Informal testing

Formative vs Summative testing

Coursework vs Examination

In summary assessment is the systematic collection and analysis of information to improve student learning.

## Evaluation

In terms of measurement evaluation is defined as ***the ability to make judgment on the basis of given information. Judging the worthiness of a programme, project or course.*** It is the use of measurement in making decisions.

### Examination

A written, oral or practical assessment, as the case may be, in accordance with general education policy.

https://www.youtube.com/watch?v=1UjiHA3g39g

https://www.youtube.com/watch?v=yRwpx__XGvQ

# MEASUREMENT AND DATA TYPES

Measurement is part of our lives and daily activities. We measure great variety of things (e.g. weight, temperature, time, distance etc). We are also being measured by doctors, teachers, supervisors etc.

## *What is measurement?*

There two common definitions of measurement, namely:

- Measurement is the act of assigning numbers or symbols to characteristics of objects (people, events, etc) according to rules.
- Measurement is the process of quantifying the degree to which someone or something possesses a given trait, i.e. quality, characteristic of feature, e.g. aggression, sociability.

### *Benefits of measurement*

A great advantage in using measurement is that one may quantify and apply the powerful tools of statistic/ mathematics to the study and describe a numbers of relationships:

i. The relation between mental ability and achievement. For example, the mental ability and achievement can be measured by assigning numbers and calculation of an index of relation between the two variables (correlation coefficient). This cannot be done through observations or descriptions.

Studies show that children who are carried at back while are infants are slow in language and emotional development. They have no opportunity to observe emotional expressions of the mothers unlike the children who are placed in front or being carried facing the mothers' faces.

One can compute correlation coefficient between the number of months the learnerwas carried at the back and scores achievement on verbal test (see hypothetical example below).

| **No. of Months Carried** | **Scores of verbal/emotional tests** |
|---|---|
| 10 | 1 |
| 9 | 2 |
| 8 | 3 |
| 7 | 4 |
| 6 | 5 |
| 5 | 7 |
| 4 | 8 |
| 3 | 9 |
| 2 | 9 |
| 1 | 10 |

ii. The relationship between the quality of food eaten and obesity; or no excise and obesity; or numbers of hour sitting watching TV and obesity. Both variables involve quantification or quantifiable measures.

| Number of Hours Watching TV per day | Weight gain per month |
|---|---|
| 12 | 10 kg |
| 10 | 9 kg |
| 9 | 8 kg |

## *What is Scale of Measurement /Levels of Measurement?*

Scales of measurement (sometimes called “level” of measurement) refers to the units in which a variable is measured: centimeters, seconds, IQ points, etc.

There are four types or basic measurement scales which can be arranged in order of information provided about the values along the scale. These measurement scales are:

- Nominal
- Ordinal

- Interval
- Ratio

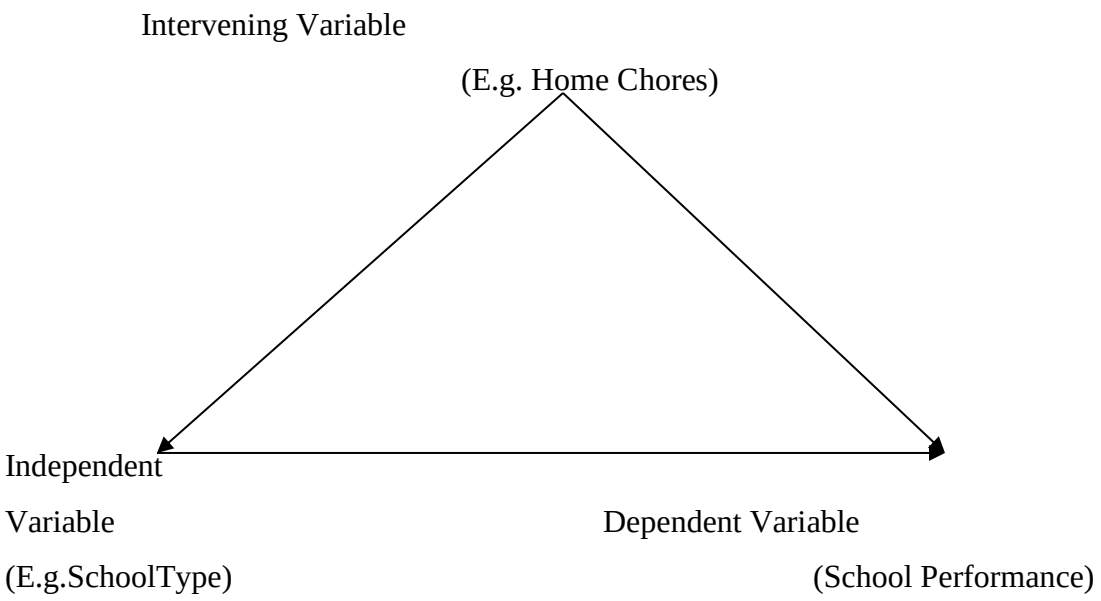
## What are variables?

In measurement we measure variables. What are variables?

- Variables are indicators, attributes or characteristics that take on different numerical values among sample of people, objects etc.
- Variables are therefore measurable entities using specific units and any nominal, ordinal, interval and ratio.

A variable can be changed or can be by itself cause change. That which can be changed is called *dependent variable* (e.g. performance in examination). The one that causes change is called *independent variable* (e.g. school type, parental income; parental level of education). The one that intervenes or comes between IV and DV is called intervening variable [eg home chores]

Examples of variables are age (5 yrs, 15 yrs, 30 yrs, 60 yrs, etc), sex (male or female), scores (20, 40, 50, etc), weight (45 kg, 65 kg, etc), income (high, medium, low), educational level (primary, secondary, university).




We measure these variables by assigning numbers.

## Nominal Scale

Which marriage option do you prefer? (Tick one choice).

1. Church wedding 27%
2. Civil wedding 4%
3. Traditional wedding 37%
4. Come-we-stay 33%

   ***Source***: Daily Nation, Saturday, July 8, 2017, p.3

- Nominal scales are the simplest form of measurement.
- A nominal scale entails the assignment of numbers or labels to objects or classes of objects. In other words, numbers are used to substitute for names.
- This scale involves identification, classification or categorization based on one or more distinguished characteristics where all objects must be placed into ***mutually exclusive and exhaustive categories.***

**Examples of Nominal**

- We classify and label people objects and places according to:
- **Sex**: male vs female and can be assigned numbers 1 = male, 2 = female.

- **Place of residence**: Nakuru or Nairobi– assign Nakuru =1, Nairobi =2
- **Jobs**: teacher, lawyer, driver, nurse; 1 = teacher, 2= lawyer, 3 = driver, 4 = nurse.
- **Size**: 1 = big, 2 = medium, 3=small
- **Colour**: 1 = Red, 2 = Black, 3 = Yellow.
- **Shape**: 1=round; 2=circular, 3=triangular, 4=rectangular
- **Numerals** on sports uniforms: The player represented by 45 is not "better" or "more than" the player represented by 32.

Use of symbols:

A=Nakuru

B=Nairobi

C= Kisumu

M= Male

F= Female

-**Political parties:** KANU, NASA, JUBILEE, KADU, SDP, KENDA

- A nominal scale is the simplest kind of scale because its rule for assigning numbers (or other labels) to objects or events is the simplest. The rule is that objects or events of the same kind get the same number and objects or events of a different kind get a different number.
- People sometimes think that a nominal scale is too primitive to be considered as proper scale.
- Has no "zero value"

## Ordinal Scale

- Ordinal scales are typically measuring of ***non-numeric concept*** like satisfaction, happiness, discomfort, birth order etc.
- Like nominal, ordinal scale permits *classification.* However, in addition to classification, rank-ordering on some characteristics is also permissible with ordinal scales.
- It ranks objects or events in order of their magnitude.
- Nominal + Rank ordering = ordinal scale
- No absolute zero point
- Measures quality rather than quantity.

**Examples of Ordinal Scale**

***How do you feel today?***

- 1. Very unhappy.
- 2. Unhappy
- 3. Ok.
- 4. Happy
- 5. Very happy

How much difference is there between very unhappy and unhappy? UNKNOWN. We cannot quantify.

***How satisfied were with EPSC 223?***

- 1. Very unsatisfied
- 2. Somewhat unsatisfied
- 3. Neutral
- 4. Somewhat satisfied
- 5. Very satisfied.

***How proud are you to be a Kenyan?***

1. Very proud.
2. Quite proud
3. Proud
4. Somehow proud
5. Not proud at all.

***Can abortion be justified?***

1. Always justified.
2. Sometimes justified.
3. Never justified.

Poor Kenyans happier than the rich, new survey shows

Society is still largely homophobic

By Vincent Achuka
vachuka@standardmedia.co.ke

Despite the tough economic times, the poorest people in Kenya (73 per cent) are living happier lives than the richest (72 per cent).

How proud are you to be Kenyan?
Nine in ten citizens are "very proud" to be Kenyan

10% Quite proud
2% Not proud
88% Very proud

Can abortion ever be justified?
One in twenty citizens says abortion can sometimes be justified

5% Sometimes
1% Always
94% Never

Study shows high uptake of medication by rape victims to protect against HIV infections at Kenyatta National Hospital. (File)

SUNDAY, JUNE 25, 2017 The Standard P. 3

**Age** (when measured using the value categories):

- Elderly
- Middle aged
- Young adult
- Adolescent
- Pre-adolescent
- Child.

**Educational attainment** (when measured by the highest degree):

- University degree
- Diploma
- Secondary certificate

- Primary certificate
- Illiterate.

**Social class standing** (when measured) by

- Rich
- Poor.

***Self-esteem***

Students' self-esteem may be scored along an ordinal scale as:

- low,
- moderate, or
- high.

**Ranking textbooks** (when measured by clarity of writing style):

Students may rank – order textbooks with respect to clarity of writing style.

- Textbook (Nominal) + Rank order
  - Book 1
  - Book 2
  - Book 3
  - Book 4
  - Book 5
- If for example, students were asked to judge 5 textbooks on clarity of writing style, they might assign:

  - The best textbook 1
  - The second textbook 2
  - The third textbook 3
  - The fourth textbook 4
  - The least favourite 5

  *You cannot say that textbook 1 is 5 times clearer than textbook 5.*

  *Kinder person assigned 1; a kinder person assigned 2.*
- ***Likert scale is an interval scale***. E.g.
  - Strongly agree

- Agree
- Fairly agree
- Disagree
- Strongly disagree

- Another characteristic of ordinal scales is that they have **<u>no absolute zero point</u>.** When using ordinal scales, no implications are made about how much greater one ranking is than another.
- Intelligence, aptitude and aggression test scores are basically ordinal. They do not indicate the amount of intelligence, aptitude and personality traits of individuals, but rather the rank-order position of individuals.

## Interval Scale

Interval scale meets all the criteria for ordinal level measurement and one additional one-the exact distance between categories of the variable are known and are equal to each other. The distance between points on the scale is fixed.

- Interval scale ***contains equal intervals between numbers.*** Interval between the numbers is equal. "Equal interval" means that the distance between the things represented by 2 and 3 is the same as the distance between the things represented by 3 and 4.
- The distances between each interval in the scale are equal.

### *Examples of Interval Scale*

***i. Measure of temperature***

***ii. Calendar Year***

- Two points next to each other on the scale, no matter whether they are high or low are separately by the same distance.
  i. Distance between $0^0$c and $10^0$c = $10^0$c
  ii. Distance between $90^0$c and $100^0$c=$10^0$c

Cf $10^0$c and $100^0$c. Does not mean $100^0$c is not 10 times hotter than something measuring $10^0$c. This is because there is no absolute zero, the zero is arbitrary.

***Why arbitrary zero?***

Zero value is taken as the point at which water freezes and $100^0$c value when water begins to boil and between these extremes values the scale is divided into a 100 equal divisions.

**100 equal divisions**

$0^0$c —/————————/————————/————————$100^0$c————————————————

- Nominal + ordinal + equal intervals = Interval scale. Each unit on the score is exactly equal to any other unit on the scale. We know exact difference between the values.
- As with ordinal scales, interval scales contain **no absolute zero point, but arbitrary zero.**
- This is a numeric scale.
  $10^0$ C + $10^0$C =$20^0$C, BUT $20^0$C is not twice hotter than$10^0$ C.

Difference between 20ºC and 30ºC is the same as the difference between 30ºC and 40ºC. Each interval is 10ºC.

- While the difference between 20ºC and 30ºC and 40ºC is the same we cannot say that 40ºC is twice as hot as 20ºC.
- Allows us to make interpretation of how warm or cold on a given day:
  - Monday = 75ºC
  - Tuesday = $65^0$C

Monday was warmer than Tuesday

- An interval scale has a zero point that does not indicate the absence of the quality. We say it has arbitrary zero, not absolute zero. **That is zero degrees C is not indicative of the complete absence of temperature. That is $0^0$c does not mean that the temperature does not exist at that point. Example Kenya's athlete whose legs were amputated because of walking frozen snow.**

- Fallacy: 80 $^0$C is twice as hot as 40 $^0$C. The statement is fallacious because the zero point on the Celsius scale has been arbitrarily set at the freezing of water.

## The case study of Kenyan athlete Cheseto amputated legs

***That is zero degrees C is not indicative of the complete absence of temperature. That is, 0$^0$c does not mean that the temperature does not exist at that point. Example: Kenya's athlete (Cheseto) whose legs were amputated because they were frozen by snow.***

On one freezing November weekend in 2011, Marko Cheseto, having finished his day's classes, decided to go for a run at around 7 p.m.

He set off on a familiar trail, and, initially all was well as he jogged into the woods. He was not wearing a jacket or gloves to beat the chilling cold. What happened next remains a mystery to date.

"I remember the first 10 minutes of my run and can't recall anything else," he told Lifestyle.

It was soon all over the news that Cheseto had gone missing for three days.

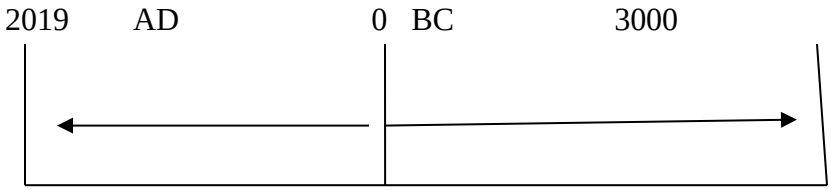
"Missing runner sparks massive search", the Anchorage Daily News headline screamed as Alaska police mounted a huge search aided by the Alaska Mountain Rescue Group's helicopters, dogs, and volunteers who included Cheseto's Kenyan colleagues and fellow students.

But after 50 hours of trawling through 15 inches of snow **in sub-zero temperatures**, police called off the search. Cheseto's colleagues on campus were devastated. They cried. Cheseto said:

"I gained consciousness at night covered deep in the snow and I did not know where I was," Cheseto, already a household name in Alaska for his running exploits, recalls. "I tried to get up but couldn't because my legs were frozen... I tried to stretch but could not."

***(b). Calendar Year***

Calendar years are another example of an interval measurement. An arbitrary 0 or 1 for hindus) was assigned when Christ was born, and time before this is given the prefix BC.




When Christ was born (starting counting point)

AD (In Latin= Anno Domini= in the year of our Lord.

Other than these examples of temperature and calendar years, interval measurement are rare.

## Ratio Scale

- Ratio scale has all the properties of nominal, ordinal and interval measurement + true zero point.
- ***Ratio = Nominal + Ordinal + Interval + Zero or absolute Point.***
- Ratio scale has a true or absolute zero point. Zero-point means that no amount of the attributes measured.
- When you are measuring the number of responses in an experiment or in observation, you are using a ratio scale. <u>Zero responses mean literary that there are no responses.</u>
- This scale has a true or absolute zero point. Zero indicates an absence of the quality measured. Zero means that no amount of the attribute measured. Zero means "none". Zero height, zero weight, zero time means no amount of these variables is present.

**Examples of Ratio:**

***i. Measures of weight***

On a measurement of weight, it is meaningful to say that an object weighing 30 kgs is three times as heavy as one weighing 10 kgs.

***ii. Measures of height***

A person who is 180 cm fall is 1.2 times taller than one who is 150 cm tall.

***iii. Measure of time***

Employees are expected to report to their work stations at 8.00 am.

Employee A reported at 9.00 am = 1 hour late

Employee B reported at 10.00 am= 2 hours late

Employee C reported at 11.00 am= 3 hours late

Employee D reported at 12.00 am= 4 hours late

Employee D arrived 4 hours late than employee A. You can count how many employees came four hour late.

***iv. Income***

Person A= earns kshs. 60,000/=

Person B= earns kshs. 20,000/=

Person A earns three times person B.

The value of zero income means no earned income.

***v. Length***

***vi. No of children in the family***

- Family A =No child
- Family B =2 children;
- Family C =3 children.

***vii. How many times a politician has been married***

- None
- Once
- Twice
- Thrice

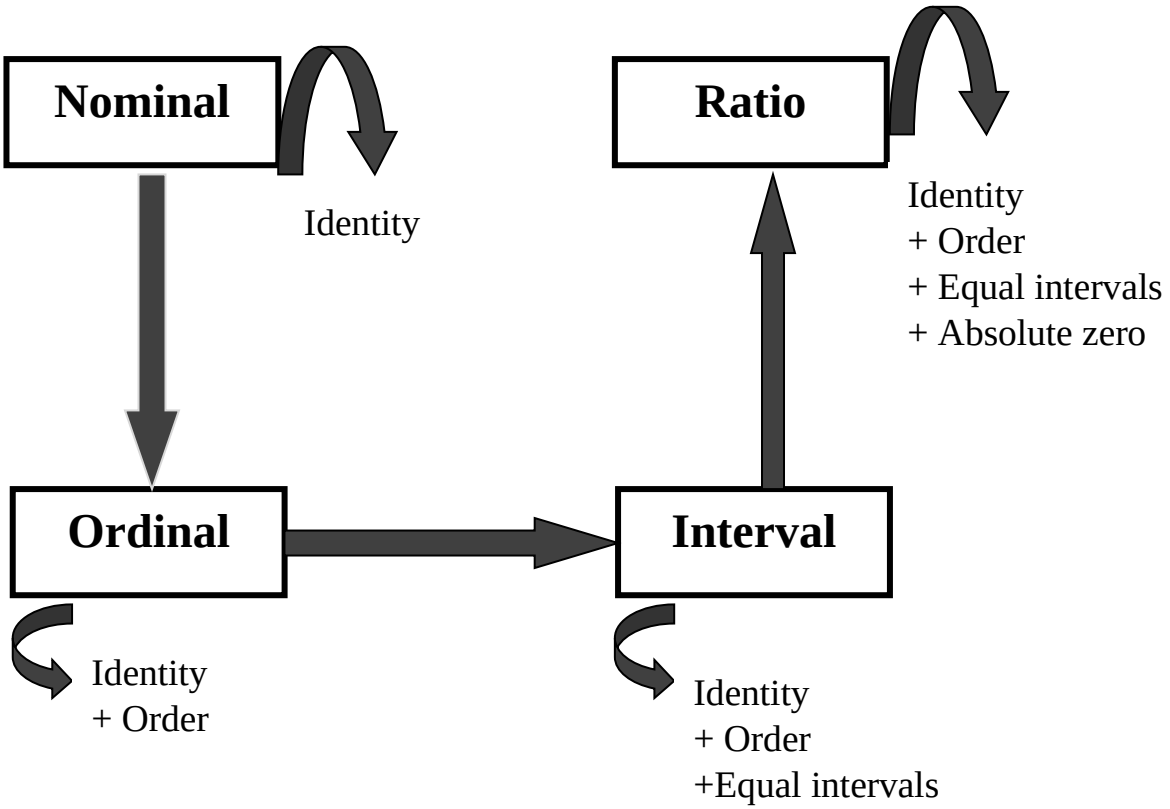
***viii. How many organizations have supported needy students at the university?***

- None
- One
- Twice
- Three

- Summary: A ratio scale is one that has a <u>meaningful zero</u> point as well as meaningful differences between the numbers on the scales.

## SUMMARY OF MEASUREMENT SCALES

- The 4 types of measurement scales are nested within one another.
  - The properties of the former are retained and additional properties are gained.
  - One can always go back to a lower measurement level from a higher one. The reverse is not generally true.
- If feasible, always aim for the highest measurement level possible



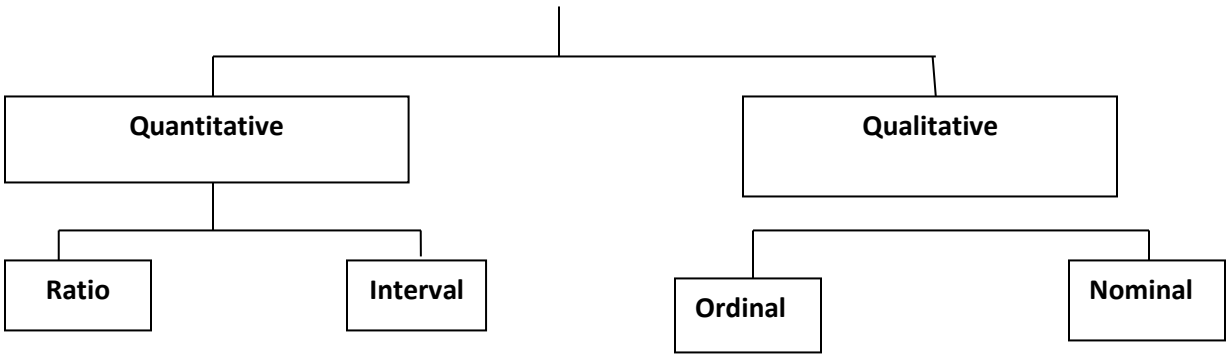

## SUMMARY OF DATA TYPES

From the measurement scales we collect two types of data, namely:

i. Quantitative data (numeric data)

ii. Qualitative data (non-numeric data)

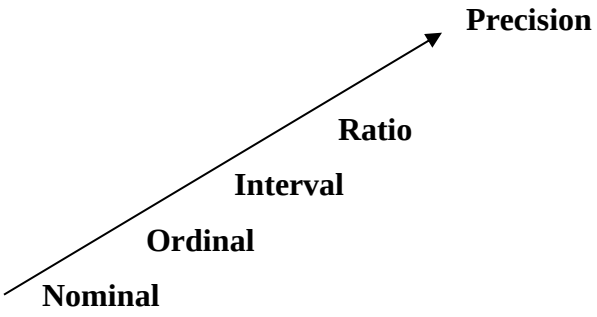
**Data Types**




# Statistics that can be computed include:

| Provides: | Nominal | Ordinal | Interval | Ratio |
|---|---|---|---|---|
| The "order" of values is known | | ✔ | ✔ | ✔ |
| "Counts," aka "Frequency of Distribution" | ✔ | ✔ | ✔ | ✔ |
| Mode | ✔ | ✔ | ✔ | ✔ |
| Median | | ✔ | ✔ | ✔ |
| Mean | | | ✔ | ✔ |
| Can quantify the difference between each value | | | ✔ | ✔ |
| Can add or subtract values | | | ✔ | ✔ |
| Can multiple and divide values | | | | ✔ |
| Has "true zero" | | | | ✔ |

**"aka"** *means "also known as"*




https://www.youtube.com/watch?v=5rUVYWfZOb8

https://www.youtube.com/watch?v=GAhniIZpqio

***Review Questions***

***a) Define the term measurement***

***b) Discuss the importance of measurement***

***c) Describe the different attributes of the different measurement scale***

***d) Discus the applicability of the different scales of measurement***

***e) With example discuss explain the concept of quantitative and qualitative data.***

**Further Reading**

1. Education Measurement and Evaluation : J.Swarupa Rani, Discovery Publishing House.

2. Measurement and Evaluation in Teaching : Norman Edward Gronlund Macmillian

3. Measurement and Assessment in Teaching : Robert L. Linn Pearson Education India.

4. Program Evaluation and performance measurement : James C. Me. David, Laura R.L. Hawthorn, Sage Publication